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# Low Power High Performance 4 bit Vedic-multiplier in 32nm

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***Abstract*— The performance of the system basically works better if the performance of the multiplier is good. In today's digital time, Multiplier is one which consumes power at the same time speed of multiplier is playing very important aspects in this. Multiplier Optimization for power and delay both will play an important role. Adders such as Ripple carry adder and carry look-ahead adder and carry skip adder are also having a role in the selection of adder units in the multiplier. In this paper, transmission gate and CMOS techniques are compared with bulk devices with different adders. Variation of 20% decrease from the operating voltage the average power, power delay product, energy-delay product decreases 76.33%,48.13% and 13.69% respectively for TG based CLA adder. Temperature increases from 25 $^0$C to 50 $^0$C, the average power, power delay product, energy delay product increases 41.42%, 65.33 % and 93.28% respectively for TG based CLA adder. Process variation from -2% to +2% to its mean value of 32nm, average power, power delay product and energy delay product decrease 68.40%, 105.86% and 85.35% for TG based CLA adder.**

***Keywords*— *Complementary Metal Oxide Semiconductor (CMOS), Transmission Gate (TG), Energy Delay Product (EDP), Power Delay Product (PDP), Ripple Carry Adder(RCA), Carry Look Ahead(CLA), Carry Skip adder(CSA), Delay, Temperature, Supply voltage***

## I. INTRODUCTION

Advances in digital systems need to emphasize on high speed and low power circuits. As today era is related to portable devices where batter power is important issues. For the longer life of battery-operated devices, one has to work on different optimization techniques. Transistor counts are the design criteria for a multiplier with different adder cells, which largely influence the design complexity of several function units in the multiplier and MAC network. The design speed is constrained by the size of the transistors, parasitic capacitance and critical path delay. A multiplier's driving skill is very significant, as often arithmetic operations are used. In the last decade, the multiplier has played a very important role in power consumption, speed and size, but there is still scope to find out the best suitable multiplier with different adders. International Technology Roadmap for semiconductors has given the power reduction in the nearest future where handheld devices will require very less power[5]. This paper presents a basic architecture of digital multipliers based on ancient Vedic mathematics Sutra (formula) called Urdhva Tiryakbhyam (Vertically and Crosswise) Sutra with three different adders. Comparison of the average power of TG and CMOS gate based 4x4 Vedic multiplier with RCA adder, CLA adder and CSA adder has been analyzed. This paper is organized in four sections. The introduction is given in the section I, Section II introduces 4-bit multiplier, Section III shows simulation results, section IV draws conclusion.

## II. SYSTEM DESIGN

### A. 4 bit Multiplier:

In this paper Average power, Delay and Energy product delay has been compared for two different adders and multipliers using TG & CMOS topology. Figure 1 shows vedic multiplier[1] replacement with RCA adder CLA[2] and CSA adder. 2-bit multiplication is shown in figure 2, figure 3 shows the 4x4 multiplier arrangement[4]. All the multiplier has been designed at the transistor level and simulated in H-spice software.

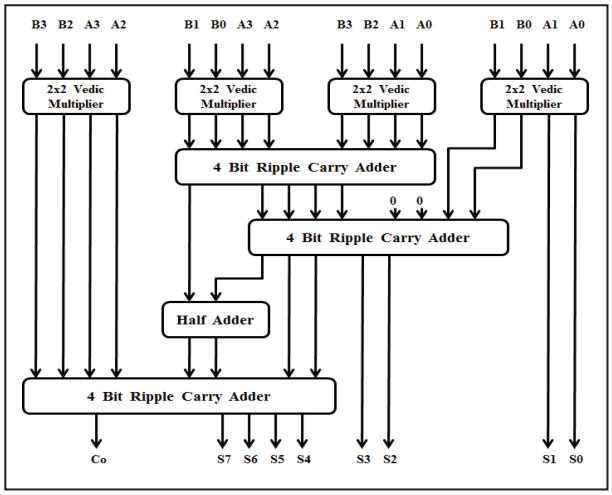



Fig. 1. Block Diagram of 4 bit Urdhva multiplier with Ripple carry adder

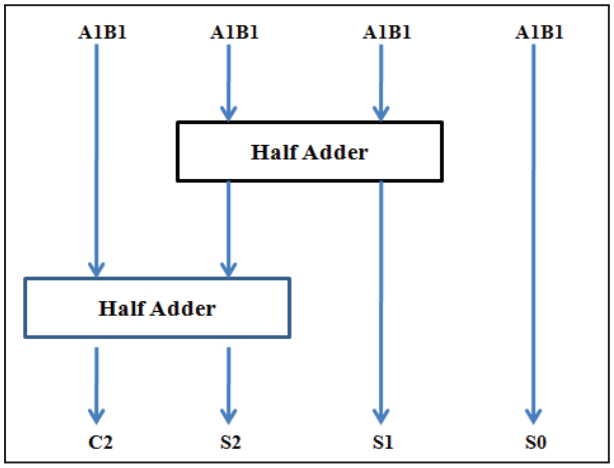



Fig. 2. 2 bit Vedic multiplication

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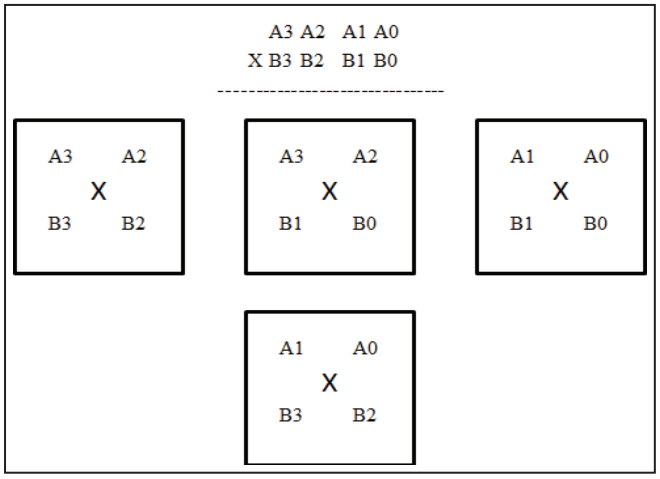



Fig. 3. 4x4 Vedic multiplication

## III. SIMULATION RESULTS

4-bit Vedic multiplier[6-7] has been designed using CMOS and TG logic style with the ripple carry, carry look ahead and carry skip adders in conventional MOSFET at 32 nm technology model. A CMOS transistor having gate length 32nm, threshold voltage 0.5088 for NMOS and threshold voltage -0.450 for PMOS, also gate length 45nm, the threshold voltage is 0.469 for NMOS and threshold voltage is -0.418V for PMOS. All these parameters has been taken from predictive technology model[5]. 4-bit Vedic multiplier using RCA adder, CLA adder and CSA adder in TG technique has been compared to find out most suitable one[8-10] As we have to try to minimize low PDP we need to get it at high frequencies at the cost of power. PDP gives the energy efficiency of the logic family. Comparison of delay, average power, PDP(Power Delay Product) and EDP(Energy Delay Product) against VDD operating voltage, temperature and process parameter has been carried out in this work.

### *A. $V_{DD}$ variation*

A graph of delay is plotted as given in fig. 4 against the operating voltage $V_{DD}$. Supply voltage variation of 20% decrease from the operating voltage, it is noted that the delay decreases. CMOS based RCA adder shows 83.97 %, CMOS based CLA adder shows 83.06 %, CMOS based CSA adder shows 83.79 %, TG based RCA adder shows 111.23 %, TG based CLA adder shows 119.16 %, TG based CSA adder shows 116.59 %. A graph of Avg. power is plotted as given in fig. 5 against the operating voltage $V_{DD}$. Supply voltage variation of 20% decrease from the operating voltage, it is noted that the average power decreases. CMOS-based RCA adder shows 77.14%, CMOS based CLA adder shows 77.21%, CMOS based CSA adder shows 77.06%, TG-based RCA adder shows 76.17%, TG based CLA adder shows 76.33%, TG based CSA adder shows 76.19%. A graph of PDP is plotted as given in fig. 6 against the operating voltage $V_{DD}$. Supply voltage variation of 20% decrease from the operating voltage, it is noted that the power delay product decreases. CMOS based RCA adder shows 57.94%, CMOS based CLA adder shows 58.29%, CMOS based CSA adder shows 57.84%, TG based RCA adder shows 49.66%, TG based CLA adder shows 48.13%,TG based CSA adder shows 48.42%.

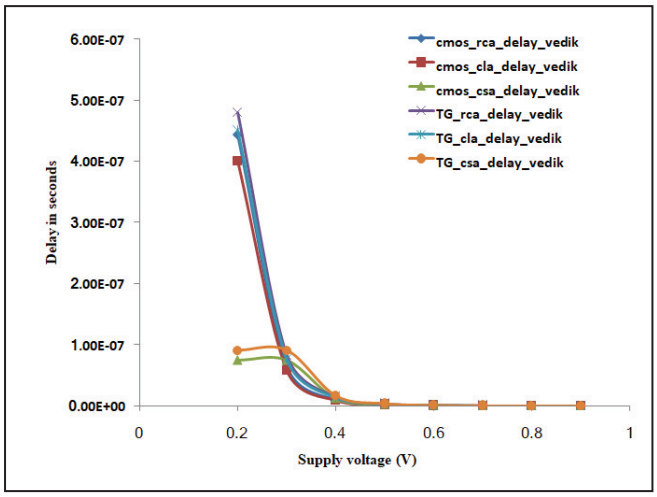



Fig. 4. Delay against $V_{DD}$

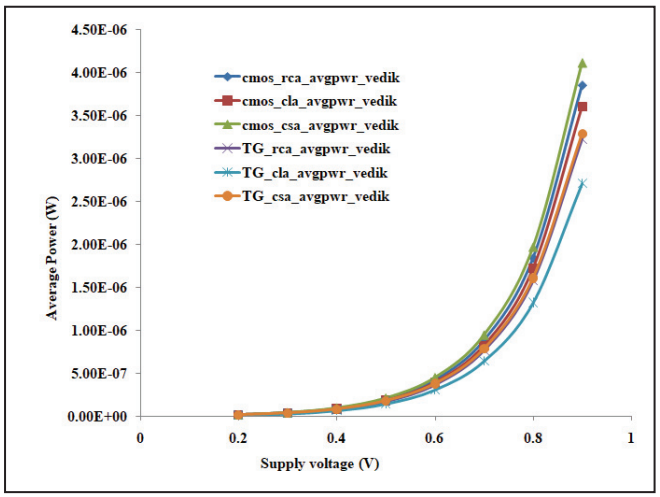



Fig. 5. Avg. power against $V_{DD}$

A graph of EDP is plotted as given in fig. 7 against the operating voltage VDD. Supply voltage variation of 20% decrease from the operating voltage, it is noted that the energy delay product decreases for CMOS based RCA adder shows 22.63%, CMOS based CLA adder shows 23.64%, CMOS based CSA adder shows 22.51%, and increases for TG based RCA adder shows 6.34%, TG based CLA adder shows 13.69%, TG based CSA adder shows 11.71%.

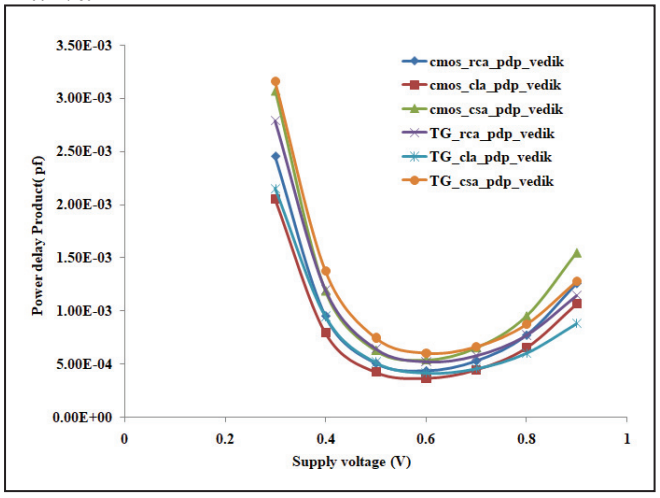



Fig. 6. PDP against $V_{DD}$



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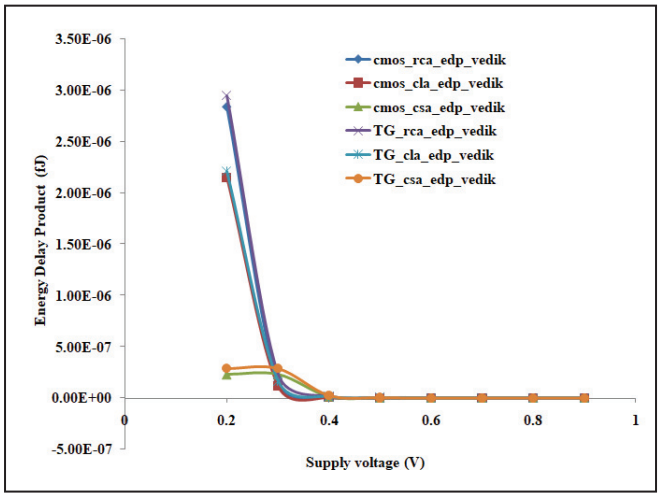



Fig. 7. EDP against $V_{DD}$

*B. Temperature variation.*

A graph of delay is plotted as given in fig. 8 against the temperature($^0$C). Temperature increases from 25$^0$C to 50$^0$C, it is noted that the delay increases, CMOS based RCA adder shows 16.09%, CMOS based CLA adder shows 16.08%, CMOS based CSA adder shows 16.11%, TG based RCA adder shows 16.52%, TG based CLA adder shows 16.90%, TG based CSA adder shows 17.35%.

A graph of Avg. power is plotted as given in fig. 9 against the temperature(0C). Temperature increases from 25$^0$C to 50$^0$C, it is noted that the average power increases. CMOS based RCA adder shows 38.90%, CMOS based CLA adder shows 38.72%, CMOS based CSA adder shows 39.04%, TG based RCA adder shows 41.92%, TG based CLA adder shows 41.42%, TG based CSA adder shows 41.85%.

A graph of PDP is plotted as given in fig. 10 against the temperature(0C). Temperature increases from 25$^0$C to 50$^0$C, it is noted that the power delay product increases. CMOS based RCA adder shows 61.24%, CMOS based CLA adder shows 61.03%, CMOS based CSA adder shows 61.45%, TG based RCA adder shows 65.36%, TG based CLA adder shows 65.33%, TG based CSA adder shows 66.46%.

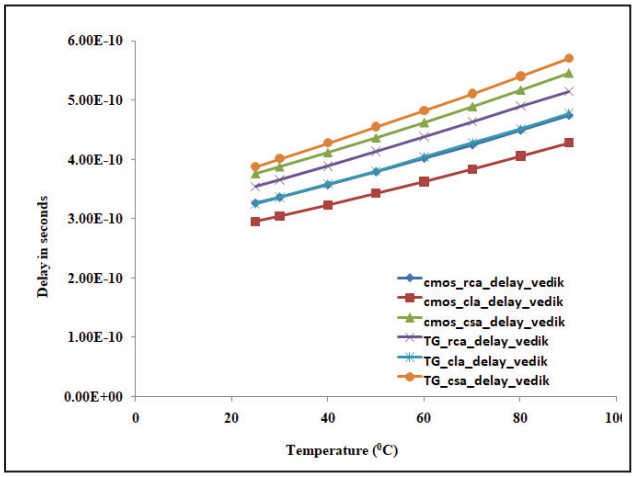



Fig. 8. Delay against temperature($^0$C)

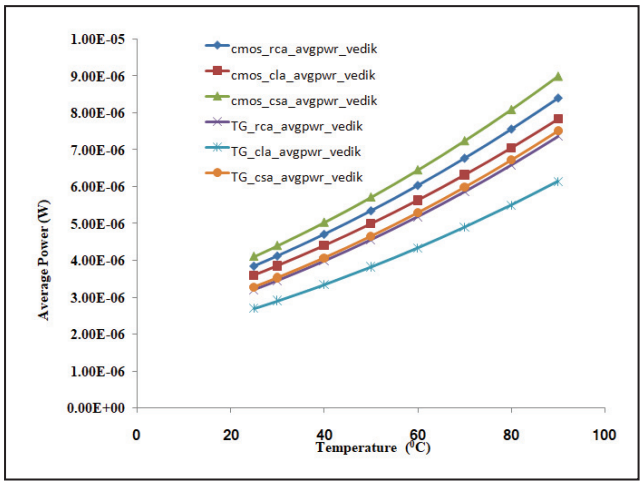



Fig. 9. Avg. power against temperature($^0$C)

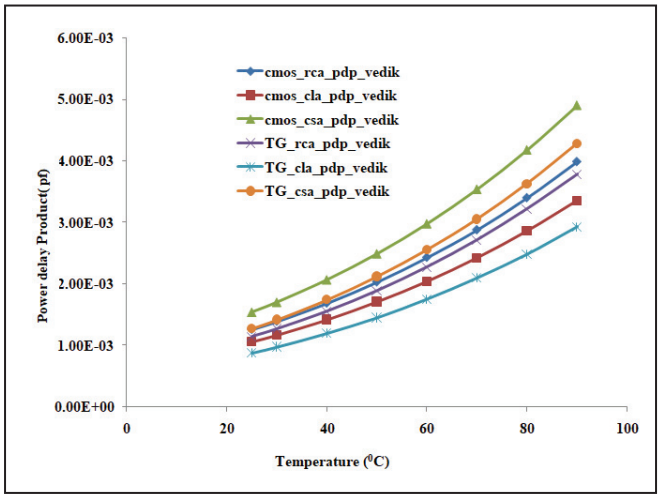



Fig. 10. PDP against temperature($^0$C)

A graph of EDP is plotted as given in fig. 11 against the temperature(0C). Temperature increases from 25 $^0$C to 50$^0$C, it is noted that the energy delay product increases for CMOS based RCA adder shows 87.19%, CMOS based CLA adder shows 86.93%, CMOS based CSA adder shows 87.46%, and increases for TG based RCA adder shows 92.68%, TG based CLA adder shows 93.28%, TG based CSA adder shows 95.34%.

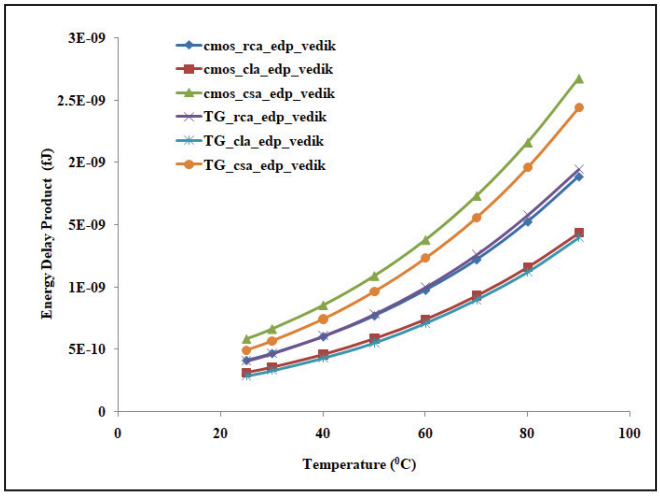



Fig. 11. EDP against temperature($^0$C)



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*C. Process variation.*

A graph of delay is plotted as given in fig. 12 against the process parameter(LG). Process variation from -2% to +2%, it is noted that the delay increases. CMOS based RCA adder shows 17.42%, CMOS based CLA adder shows 11.19%, CMOS based CSA adder shows 11.10%, TG based RCA adder shows 17.18%, TG based CLA adder shows 17.62%, TG based CSA adder shows 18.04%.

A graph of Avg. power is plotted as given in fig. 13 against the process parameter(LG). Process variation from -2% to +2%, it is noted that the average power decrease. CMOS based RCA adder shows 69.23%, CMOS based CLA adder shows 66.05%, CMOS based CSA adder shows 64.70%, TG based RCA adder shows 68.17%, TG based CLA adder shows 68.40%, TG based CSA adder shows 68.20%.

A graph of PDP is plotted as given in fig. 14 against the process parameter(LG). Process variation from -2% to +2%, it is noted that the power delay product decrease. CMOS based RCA adder 109.36%, CMOS based CLA adder shows 106.46%, CMOS based CSA adder shows 102.26%, TG based RCA adder shows 105.54%, TG based CLA adder shows 105.86%, TG based CSA adder shows 104.51%.

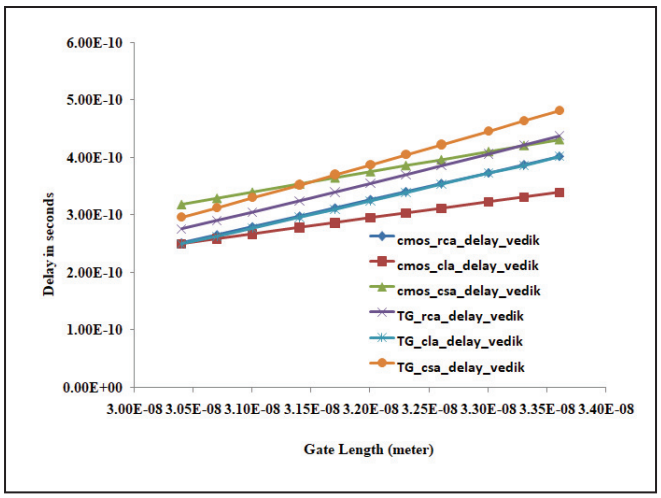



Fig. 12. Delay against process parameter($L_G$)

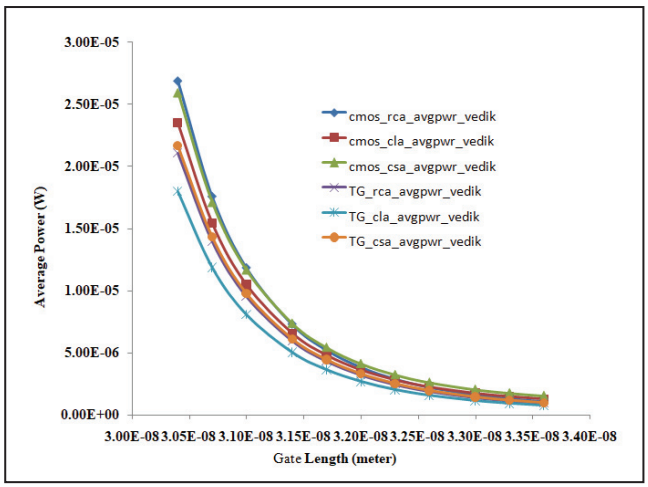



Fig. 13. Avg. power against process parameter($L_G$)

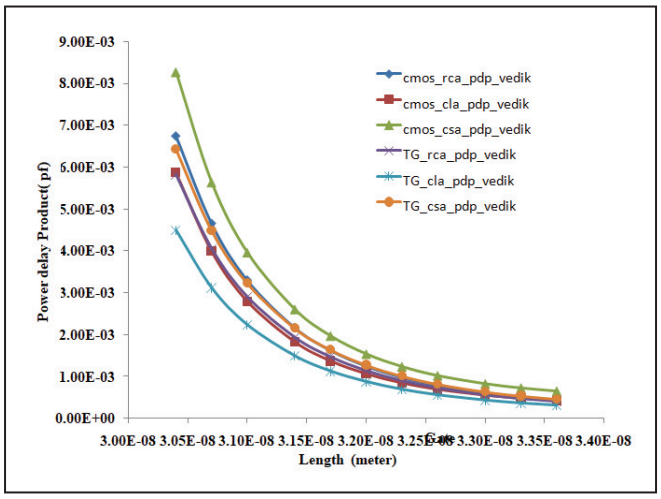



Fig. 14. PDP against process parameter($L_G$)

A graph of EDP is plotted as given in fig. 15 against the process parameter(LG). Process variation from -2% to +2%, it is noted that the energy delay product decrease for CMOS based RCA adder shows 88.74%, CMOS based CLA adder shows 93.10%, CMOS based CSA adder shows 89.11%, TG based RCA adder shows 85.57%, TG based CLA adder shows 85.35%, TG based CSA adder shows 83.48%.

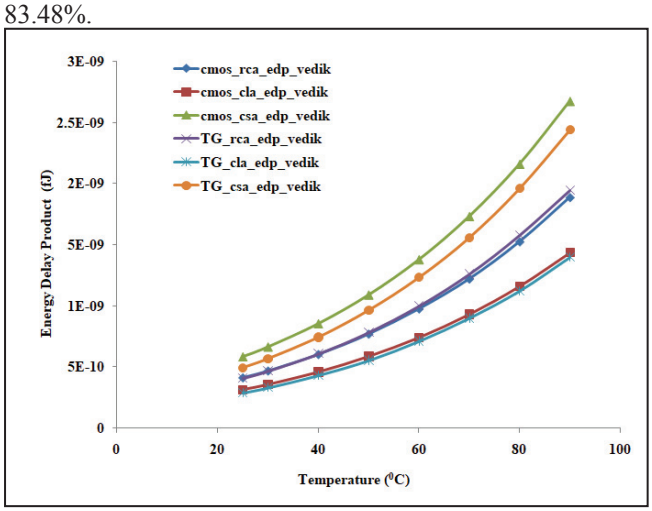



Fig. 15. EDP against process parameter($L_G$)

The results that were carried out at room temperature 250C and VDD with all the foundry parameters of the MOSFET file taken from ITRS are depicted in Table 1.

TABLE I. READING CARRIED OUT AT SUPPLY VOLTAGE 0.9V AND ROOM TEMPERATURE $25^0$C.

| | CMOS | | | TG | | |
|---|---|---|---|---|---|---|
| | RCA | CLA | CSA | RCA | CLA | CSA |
| Delay (ps) | 327 | 295 | 376 | 355 | 325 | 388 |
| Avg. Power (Micro W) | 3.85 | 3.60 | 4.11 | 3.22 | 2.71 | 3.29 |
| PDP ( fj ) | 1.26 | 1.06 | 1.55 | 1.14 | 0.879 | 1.27 |
| EDP ( fj ) | 4.11 e-10 | 3.14 e-10 | 5.8 e-10 | 4.06 e-10 | 2.85 e-10 | 4.94 e-10 |



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## IV. CONCLUSION

4-bit Vedic multiplier using RCA adder, CLA adder and CSA adder in TG technique has been compared to find out most suitable one It is to optimize the PDP, the work has been carried out at high frequency with low power devices irrespective of the cost of power. PDP gives the energy efficiency of the logic family. Comparison of delay, average power, PDP and EDP as a variation in Supply Voltage, Temperature and process parameters has been carried out in this work.

For the variation of 20% decrease from the operating voltage, it is noted that the delay increases 83.06 % for CMOS based CLA adder, Whereas the average power decreases 76.33%, power delay product decreases 48.13% and energy delay product decreases for 13.69% for TG based CLA adder. Temperature increases from $25^0$C to $50^0$C, it is noted that the delay increases 16.08% for CMOS based CLA, average power increases 41.42%, power delay product increases 65.33 % and the energy delay product increases 93.28% for TG based CLA adder. Process variation from -2% to +2%, it is noted that the delay increases 11.19% for CMOS based CLA adder, average power decrease 68.40%, power delay product decrease 105.86% and energy delay product decrease 85.35% for TG based CLA adder.

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